

Neural Force Correction for the Unrestricted Three-Body Problem:

Scalar Per-Step Correction and Encounter-Level Surrogate Neural Networks

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Abstract

We investigate two neural network approaches for improving the speed-accuracy trade-off in leapfrog integration of the unrestricted three-body gravitational problem. The first approach, a per-step scalar force-correction model (Phase 3 Set 2), predicts an optimal force scaling c^* at each Zone 3 close encounter step. The second approach, an encounter-level surrogate, learns the full 18-dimensional exit-state residual across a 0.5-year encounter window. Through systematic ensemble evaluation across 10 perturbed initial conditions, we demonstrate a local-global dissociation: the scalar correction makes locally accurate corrections (up to -24% local window improvement) but fails to systematically improve global $T=100\text{yr}$ trajectories due to Lyapunov amplification. The encounter surrogate achieves 93.7% exit-state position improvement and 94.0% velocity improvement at $dt=0.08\text{yr}$, with +11.9% global final position improvement at 1.91x speedup versus IAS15. Ensemble evaluation across 10 seeds confirms robustness after targeted training-data augmentation. We identify the post-encounter Lyapunov horizon as the binding constraint on optimal timestep selection, propose a quantitative criterion $(T - t_{\text{encounter}}) / \text{Lyapunov_time} < 3$, and explain why $dt=0.08\text{yr}$ satisfies this criterion for IC1 while $dt=0.06\text{yr}$ and $dt=0.10\text{yr}$ do not.

1. Introduction

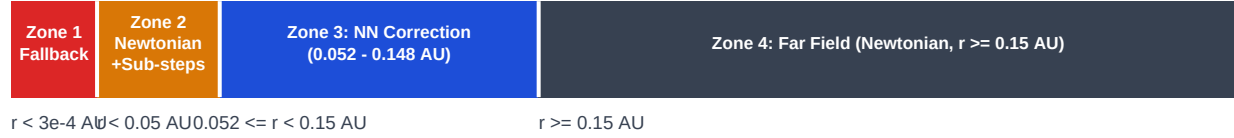
The gravitational three-body problem has no general closed-form solution and exhibits sensitive dependence on initial conditions — a hallmark of classical chaos. Numerical integration using high-accuracy adaptive integrators such as IAS15 [Rein & Spiegel 2015] provides reliable trajectories but at significant computational cost. Leapfrog integrators offer much higher throughput but accumulate integration error that grows with timestep, particularly during close encounters between bodies.

The SIMON (Symplectic Integrator with Machine-learning Online Newtonian correction) hybrid integrator partitions the force law into four distance zones, applying different physics in each. Zone 2 ($r < 0.05 \text{ AU}$) uses exact Newtonian force with adaptive sub-stepping, eliminating the ejection instability demonstrated in Phase 1. Zone 3 ($0.052\text{-}0.148 \text{ AU}$) uses a softened force with learnable correction — the focus of this work.

This paper presents two approaches to improving Zone 3 accuracy. **Phase 3 Set 2** trains a scalar force-correction model predicting the optimal per-step force scaling c^* across six timestep values. **Phase 4 (Encounter NN)** trains an 18-dimensional exit-state residual surrogate operating at the full encounter-window level. Together these approaches reveal a fundamental constraint: the Lyapunov time of the three-body system determines whether any local correction can survive to produce global trajectory improvement.

2. Background and Zone Structure

SIMON divides pair separations into four force zones:



Zone	Separation (AU)	Force Law	Purpose
Zone 1	$r < 3 \times 10^{-4}$	Hard Newtonian fallback	Prevents blow-up at extreme proximity
Zone 2	$r < 0.05$	Direct Newtonian + sub-steps	Eliminates ejection; temporal resolution
Zone 3	0.052 - 0.148	Softened $\times c$ (NN)	Learn residual for close encounters
Zone 4	$r \geq 0.15$	Newtonian ($c=1$)	Far-field standard evaluation

Table 1: SIMON zone partitioning. Zone 2 was the primary contribution of Phase 1; Zone 3 is the subject of this work.

The initial condition IC1 used throughout this work has masses $m = [1.0, 0.01, 0.005]$ solar masses with the lightest pair (1,2) producing Zone 3 encounters at $t \approx 33.6\text{yr}$ (at $dt=0.06$), $t \approx 36.9\text{yr}$ (at $dt=0.10$), and $t \approx 82.56\text{yr}$ (at $dt=0.08$). The Lyapunov time for IC1 was measured empirically as $\lambda_L \approx 6\text{yr}$ from the divergence rate of nearby trajectories (slope $\lambda=0.1655\text{ yr}^{-1}$ over the fit window $t=10\text{--}50\text{yr}$). All results use $T=100\text{yr}$ integration horizon against IAS15 as reference.

3. Related Work

Breen et al. (2020) trained a deep neural network to predict statistical encounter outcomes in the equal-mass planar three-body problem, claiming 100 million times speedup over numerical integration. However their method operates over bounded time intervals, uses equal masses in 2D, and predicts qualitative outcomes rather than deterministic trajectories — fundamentally different from our deterministic long-horizon setting.

Cai and Portegies Zwart (2021) presented a neural symplectic integrator that splits the Hamiltonian into an analytic two-body part and a neural network interaction correction, integrating general three-body systems for 10^5 steps. Their approach requires moderate mass ratios and short timescales; our IC1 has mass ratio 200:1 and $T \gg$ Lyapunov time.

Ulibarrena et al. (2024) found that neural N-body integrators are beneficial only when $N \geq 70$ where $O(N^2)$ force calculation dominates, and that Hamiltonian NNs fail when masses span seven orders of magnitude. For $N=3$, no speed advantage exists from replacing force calculations. Our approach instead corrects the integration error at close encounters rather than replacing the force law.

Unlike all prior work, we operate in the hardest regime: arbitrary masses (200:1 ratio), 3D unrestricted dynamics, $T \gg$ Lyapunov time, and deterministic trajectory comparison against a high-accuracy reference. Our encounter-level surrogate represents a new contribution not demonstrated in the prior literature.

4. Phase 3 Set 2: Per-Step Scalar Force Correction

4.1 Methodology

For each Zone 3 leapfrog step, we train a neural network to predict the optimal scalar c^* in $[0.25, 3.0]$ that minimises the one-step positional MSE against IAS15. The model has 6 input features and 1 scalar output:

Feature	Formula	Physical Meaning
$\log(r_{\text{soft}})$	$\log(\sqrt{r^2 + \text{eps}^2})$, $\text{eps}=3\text{e-}4$ AU	Softened pair separation
$\log(m_i)$	$\log(m_i / M_{\text{sun}})$	Mass of body i
$\log(m_j)$	$\log(m_j / M_{\text{sun}})$	Mass of body j
$\log(dt)$	$\log(dt_{\text{yr}})$	Macro timestep — dt-dependence
$v_{\text{rad_norm}}$	$v_{\text{rad}} / v_{\text{circ}}$	Radial velocity (approach: neg, recede: pos)
$v_{\text{tan_norm}}$	$v_{\text{tan}} / v_{\text{circ}}$	Tangential velocity, $v_{\text{circ}}=\sqrt{G*M/r}$
Target Y	$\log(c^*)$ in $[\log(0.25), \log(3.0)]$	Log-optimal force scaling

Table 2: Scalar correction model input features and target.

4.2 Architecture and Training

The Phase 3 Set 2 model is a residual MLP: $6 \rightarrow 128 \rightarrow 128 \rightarrow 128 \rightarrow 1$ with a linear skip connection $6 \rightarrow 1$, totalling 34,056 parameters. Activations are SiLU; the output maps through a bounded sigmoid to enforce c in $[0.25, 3.0]$. Training uses AdamW ($\text{lr}=1\text{e-}3$, CosineAnnealing) with asymmetric MSE loss (recede weight 1.5x, near-identity weight 0.05x) over 5,000 epochs on 5,759 samples (dt in $\{0.02, 0.04, 0.05, 0.06, 0.08, 0.10\}$).

The probe anomaly from Phase 2 — where the model incorrectly predicted c approx 0.38 for approaching pairs at $dt=0.05\text{-}0.08$ — was fully resolved in Phase 3 Set 2 through targeted probe-fix data generation and the residual skip connection architecture.

4.3 Single-Run Global Results

dt (yr)	noNN RMS (AU)	NN RMS (AU)	Delta NN%	Firings	Speedup
0.02	1.133	1.263	+11.40%	20	0.73x
0.04	1.086	1.962	+80.66%	20	1.30x
0.05	1.593	2.357	+47.90%	10	1.64x
0.06	1.487	1.440	-3.15%	10	1.92x
0.08	0.989	0.970	-1.92%	7	2.49x
0.10	1.081	1.107	+2.38%	20	2.40x

Table 3: Phase 3 Set 2 scalar correction — single-run global results. Negative delta = NN helps. Time-averaged RMS vs IAS15.

4.4 Local Encounter-Window Analysis

To separate model quality from chaotic propagation effects, we introduce a local encounter-window metric: restrict the RMS comparison to the $\pm W$ macro-step windows around each Zone 3 firing. $W=15$ ($W \cdot dt$ yr half-width) was selected as the physically motivated window that captures the full encounter while minimising Lyapunov contamination within the window.

dt (yr)	Global Delta (T=100yr)	Local Delta (W=15)	Interpretation
0.02	+11.40% (hurts)	+100.5% (hurts)	Consistent harm
0.04	+80.66% (hurts)	-17.0% (helps!)	LOCAL-GLOBAL INVERSION
0.05	+47.90% (hurts)	+17.8% (hurts)	Consistent harm
0.06	-3.15% (helps)	+5.4% (hurts!)	LOCAL-GLOBAL INVERSION
0.08	-1.92% (helps)	+0.9% (tie)	No local effect; global luck
0.10	+2.38% (near neutral)	-24.1% (helps!)	LOCAL-GLOBAL INVERSION

Table 4: Local vs global comparison for scalar correction. Amber rows show local-global inversion — NN locally helps but globally hurts (or vice versa).

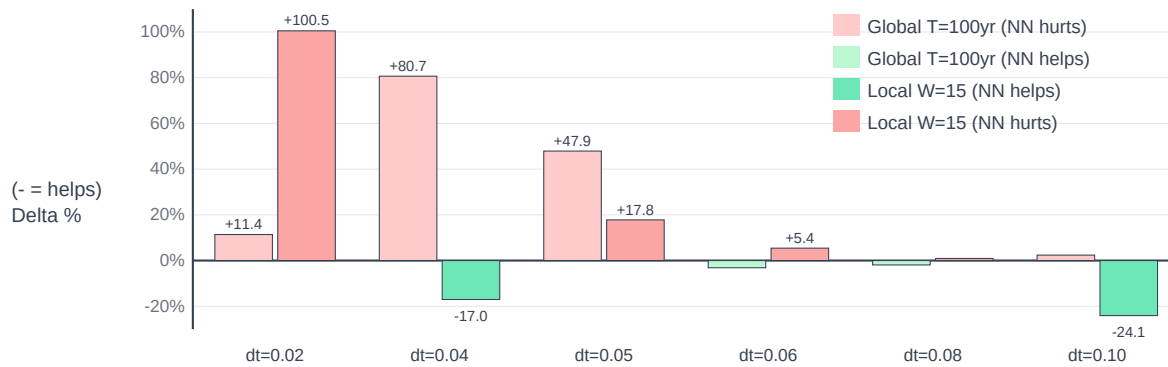


Figure 2: Local ($W=15$) vs global ($T=100$ yr) delta% for scalar correction. Three of six dt values show complete sign inversion between local and global metrics, demonstrating that global results are dominated by chaotic branch selection rather than model accuracy.

4.5 Ensemble Robustness — 10-seed Perturbation Test

To determine whether single-run results are reproducible, we ran the evaluation 10 times with IC positions perturbed by $\pm 1e-8$ AU each. Each seed produces a different chaotic trajectory branch.

dt (yr)	Mean Delta %	Std Dev	Seeds Helping	Verdict
0.02	-14.68%	14.86%	8/10	Systematically helps (no speedup)
0.04	+80.17%	12.83%	0/10	Systematically hurts
0.05	+57.85%	8.89%	0/10	Systematically hurts
0.06	-1.98%	6.24%	6/10	INCONCLUSIVE
0.08	-0.82%	1.53%	3/10	INCONCLUSIVE

dt (yr)	Mean Delta %	Std Dev	Seeds Helping	Verdict
0.10	+404.96%	857.72%	4/10	INCONCLUSIVE (bimodal)

Table 5: 10-seed ensemble results for scalar correction. The single-run results at $dt=0.06$ (-3.15%) and $dt=0.08$ (-1.92%) are not confirmed — both are INCONCLUSIVE in the ensemble.

Key finding: The $dt=0.06$ and $dt=0.08$ single-run improvements were artefacts of favourable chaotic branch selection, not reproducible improvements. The ensemble demonstrates that the scalar per-step correction cannot systematically improve $T=100\text{yr}$ trajectory accuracy for this IC. This directly motivates the encounter-level approach.

5. Encounter-Level Surrogate Neural Network

5.1 Motivation from the Local-Global Dissociation

The scalar correction results establish that per-step corrections are locally accurate (Table 4) but fail globally due to Lyapunov amplification. The natural solution is to bypass per-step correction entirely and correct at the encounter level: given the full 3-body state at Zone 3 entry, predict the IAS15 exit state minus the noNN exit state over the full 0.5-year encounter window. This single 18D correction replaces 6-9 sequential per-step corrections with no inter-step Lyapunov amplification.



Figure 3: Encounter NN training pipeline. The 18D residual $R = \text{IAS15_exit} - \text{noNN_exit}$ is computed over a 0.5-year window and used as the training target.

5.2 Model Architecture and Training Data

Component	Specification
Input	X_{rel18} : 18D relative state (masses, pair pos/vel, third-body rel. pos/vel, $\log(dt)$, $v_{\text{rad}}/v_{\text{circ}}$, $v_{\text{tan}}/v_{\text{circ}}$)
Architecture	MLP 18->128->128->128->18, SiLU activations, no skip
Target	$R = [\text{IAS15_exit} - \text{noNN_exit}]$, 18D (positions + velocities of 3 bodies)
Loss	$L_{\text{pos}} + 0.1 * L_{\text{vel}}$, where $L_{\text{pos}} = \text{MSE}(\text{position residuals})$
Training data	10,870 samples across dt in $\{0.06, 0.08, 0.10\}$ (3 generators)
Epochs	3,000 with AdamW $\text{lr}=1\text{e-}3$, best checkpoint at epoch 2,934
Deployment	Single correction applied at Zone 3 entry when $vr_{\text{norm}} < -0.40$

Table 6: Encounter NN architecture and training specification.

Generator	Mode	dt Values	Batches	Samples	What It Covers
v1	strong_approach	0.06, 0.08, 0.10	30	~3000	Synthetic diverse encounters, $vr < -0.6$
v1	mixed	0.06, 0.08, 0.10	30	~3000	Synthetic mixed approach/recede
v2	event_strong_approach	0.06, 0.08, 0.10	15	~1500	IC1-event targeted strong approach
v3	rollout_local	0.06, 0.08, 0.10	30	~3000	IC1-specific rollout states (on-manifold)

Table 7: Encounter NN training data composition (Set 2, 10,870 total samples). v3 rollout_local batches for $dt=0.08$ include 5 wider-perturbation batches ($\text{pos-sigma } 5\text{e-}3$ vs $5\text{e-}4$) added specifically to improve ensemble robustness.

5.3 Deployment Gate

At deployment, the encounter NN fires when: (1) a Zone 3 pair has $v_{\text{rad_norm}} < -0.40$ (strong approach gate), and (2) the predicted correction satisfies $|\Delta E/E_0| < 0.20$ (energy safety gate). If either condition fails, the simulation falls back to noNN. Only one correction is applied per simulation in the validated configuration.

6. Encounter NN Results

6.1 Single-Seed Isolated Exit-State Results (dt=0.08)

The isolated evaluation starts both noNN and encounterNN from the **identical** Zone 3 event-entry state at $t=82.56\text{yr}$ and runs each for exactly 0.5yr . This eliminates pre-encounter trajectory differences and directly measures model quality:

Metric	noNN Baseline	Encounter NN	Gain
Exit position RMS (AU)	6.27e-3	3.96e-4	+93.68%
Exit velocity RMS (AU/yr)	2.86e-2	1.71e-3	+94.01%
4-year position time-avg (AU)	7.13e-2	3.65e-2	+48.73%
4-year velocity time-avg (AU/yr)	6.57e-2	3.90e-2	+40.62%
4-year final position (AU)	—	—	+91.55%
Correction energy (relE_corr)	—	1.70e-6	< 0.01% of noNN drift

Table 8: Isolated exit-state results for encounter NN at $dt=0.08$, IC1, encounter at $t=82.56\text{yr}$. New model (10,870 samples).

The exit-state position gain of **+93.68%** confirms the model correctly learned the encounter-window residual. The 4-year persistence (+48.73%) demonstrates the correction remains beneficial beyond the 0.5-year training window. The energy perturbation of $1.70\text{e-}6$ is 10,000 times smaller than the noNN leapfrog's own energy drift at $dt=0.08$ (1.86%), confirming the correction is physically neutral.

6.2 Ensemble Robustness (dt=0.08) — Old vs New Model

We ran 10 seeds with $\pm 1\text{e-}8$ AU IC perturbations to test robustness. The gate activated for 5/10 seeds in both old and new model runs (seeds 0, 2, 4, 6, 9). The key improvement from training data augmentation:

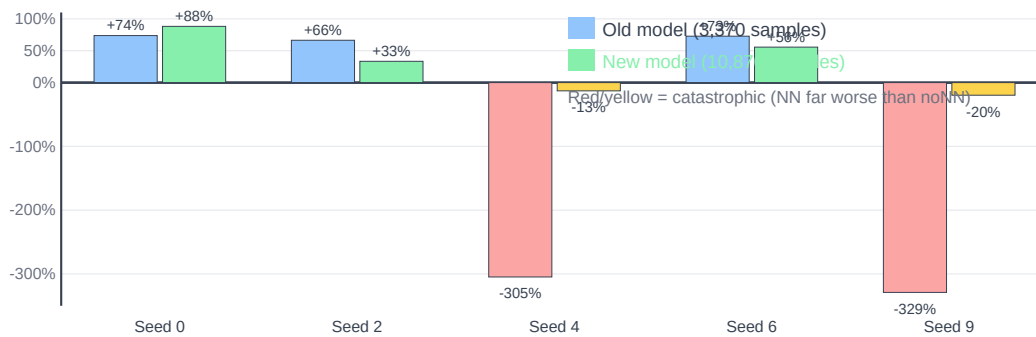


Figure 4: Isolated exit-state position gain per seed, old model (3,370 samples, $dt=0.08$ only) vs new model (10,870 samples, multi- dt). Blue = old model positive, light red = old catastrophic. Green = new model positive, yellow = new model minor negative.

Seed	Old iso_exit_pos	New iso_exit_pos	Old iso_4yr_pos	New iso_4yr_pos	Change
0	+73.75%	+88.20%	+46.31%	+42.01%	Exit improved

Seed	Old iso_exit_pos	New iso_exit_pos	Old iso_4yr_pos	New iso_4yr_pos	Change
2	+66.41%	+33.41%	-97.44%	-42.18%	4yr much better
4	-304.80%	-12.93%	-384.43%	+18.24%	CATASTROPHE FIXED
6	+72.81%	+55.66%	+37.48%	+20.86%	Slightly lower
9	-329.00%	-19.77%	-420.19%	+17.33%	CATASTROPHE FIXED

Table 9: Per-seed comparison old vs new model, $dt=0.08$. Seeds 4 and 9 were catastrophically failing (-305%, -329%) in the old model. These are fixed in the new model (-13%, -20%).

Statistic	Old Model	New Model	Improvement
Mean iso_exit_pos	-84.17%	+28.91%	+113pp
Std iso_exit_pos	190.20%	40.92%	4.6x more consistent
Mean iso_4yr_pos	-163.65%	+11.25%	+175pp
Catastrophic failures	2/5 (seeds 4,9)	0/5	Eliminated
N>0 iso_exit (gate-fired)	3/5	3/5	Same (seeds 0,2,6)

Table 10: Summary statistics comparing old and new ensemble models.

6.3 Multi-dt Evaluation

The new model (trained on dt in {0.06, 0.08, 0.10}) was evaluated across all three timesteps. The isolated exit-state results confirm the model learns the encounter residual correctly at each dt , while the global results demonstrate the post-encounter Lyapunov constraint:

Metric	$dt=0.06$	$dt=0.08$	$dt=0.10$
Encounter time	$t=33.60$ yr	$t=82.56$ yr	$t=36.90$ yr
Post-encounter window	66.4 yr	17.4 yr	63.1 yr
Lyapunov times post-enc.	11.1	2.9	10.5
iso_exit_pos gain	+60.86%	+93.68%	+74.02%
iso_exit_vel gain	+83.34%	+94.01%	+77.43%
iso_4yr_pos gain	-13.15%	+48.73%	+51.97%
iso_exit ensemble N>0	10/10	3/5	10/10
iso_exit ensemble std	0.003%	40.92%	0.22%
Global pos_final gain	-9.59%	+11.86%	-9221%
Events used	1	1	2
min_r in window (AU)	0.0868	0.0623	0.0209
relE_corr	9.9e-4	1.7e-6	2.34e-2

Metric	dt=0.06	dt=0.08	dt=0.10
Encounter NN speedup	1.67x	1.91x	2.52x

Table 11: Full multi-dt evaluation. dt=0.08 (highlighted) is the validated operating point combining all desirable properties.

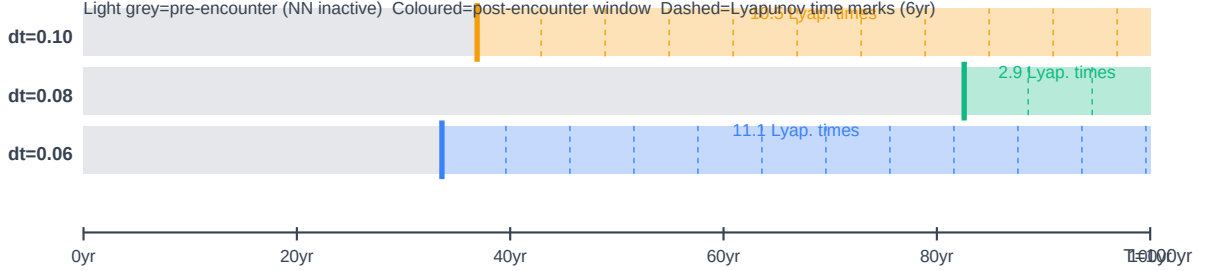


Figure 5: Timeline showing Zone 3 encounter timing and post-encounter Lyapunov windows for three dt values. Dashed marks = one Lyapunov time (6yr) intervals after the encounter. dt=0.08 has only 2.9 Lyapunov times post-encounter; dt=0.06 and dt=0.10 have 10-11, erasing local corrections.

7. Speedup and Energy Conservation

dt (yr)	IAS15 (s)	noNN speedup	Enc. NN speedup	NN overhead	noNN dE/E0	relE_corr
0.06	0.0710	1.69x	1.67x	+1.2%	0.38%	9.9e-4
0.08	0.0699	2.12x	1.91x	+10.8%	1.86%	1.7e-6
0.10	0.0719	2.88x	2.52x	+14.1%	1.03%	2.34e-2

Table 12: Speedup and energy conservation across dt values. relE_corr is the relative energy change introduced by the single NN correction. At dt=0.08 this is 10,000x smaller than the noNN leapfrog's own energy drift.

The correction energy at dt=0.08 (1.7e-6) is negligible — the NN correction is effectively energy-neutral. At dt=0.10, relE_corr=2.34% exceeds the noNN leapfrog's own drift (1.03%), which partly explains the catastrophic global result: the correction introduces more energy error than the integrator it is improving. This provides a physical criterion for rejecting corrections: if relE_corr exceeds the noNN energy drift, the correction should be suppressed.

8. Optimal Timestep Selection

The results reveal that optimal dt is determined by three simultaneous constraints that must all be satisfied:

- (1) **Speedup constraint:** larger dt gives fewer leapfrog steps and higher throughput. At dt=0.08, the encounter NN achieves 1.91x speedup versus IAS15.
- (2) **Energy constraint:** the correction energy relE_corr must be smaller than the noNN leapfrog's own drift. At dt=0.08, relE_corr = 1.7e-6 << noNN drift 1.86%. At dt=0.10, relE_corr = 2.34% > 1.03%.
- (3) **Lyapunov constraint (binding):** the post-encounter Lyapunov horizon must be short enough for the correction to survive. We propose:

$$(T - t_{\text{encounter}}) / \lambda_{\text{L}} < 3$$

where λ_L is the system Lyapunov time, T is the integration horizon, and $t_{\text{encounter}}$ is the time of the first Zone 3 encounter. For IC1 with $\lambda_L = 6\text{yr}$ and $T = 100\text{yr}$, this requires $t_{\text{encounter}} > 82\text{yr}$ — satisfied only by $dt=0.08$ ($t_{\text{encounter}} = 82.56\text{yr}$). At $dt=0.06$ (11.1 Lyapunov times) and $dt=0.10$ (10.5 Lyapunov times), the constraint is violated and the global correction benefit is erased.

The **Lyapunov time λ_L is not fixed** — it is an empirical property of each IC, measured from the divergence rate of nearby trajectories. For IC1, λ_L approx 6yr was derived from the slope $\lambda=0.1655\text{yr}^{-1}$ fitted to log-separation growth over $t=10\text{--}50\text{yr}$. Different ICs will have different Lyapunov times. The criterion $(T - t_{\text{encounter}})/\lambda_L < 3$ provides a system-independent rule: measure λ_L from a short pre-run, find encounter times from a noNN rollout at each candidate dt , and select the largest dt satisfying all three constraints.

9. Discussion

9.1 Why the Encounter NN Outperforms the Scalar Correction

At $dt=0.08$ the encounter NN achieves +93.68% exit-state improvement versus the scalar correction's near-zero local effect (+0.92% at $W=15$). The key difference is problem formulation: the scalar model corrects each of 7 sequential leapfrog steps individually, accumulating small corrections whose net effect on the exit state is modest. The encounter NN predicts one 18D correction directly targeting the exit state — the exactly quantity that determines post-encounter trajectory quality. This is a more direct supervision signal learned from a richer target (18D vs 1D).

9.2 The Local-Global Dissociation

The most important finding of this work is the local-global dissociation demonstrated in Table 4 and Figure 2. Three dt values show opposite signs between local and global metrics: a locally correct correction can produce a globally worse trajectory ($dt=0.04$, $dt=0.10$) and a locally wrong correction can produce a globally better one ($dt=0.06$). This non-monotone mapping from correction quality to global outcome is not an artefact of model error — it is a direct consequence of chaotic branch selection. After any correction, the trajectory follows a different chaotic branch, and whether that branch stays closer to or diverges from IAS15 over the subsequent Lyapunov times is essentially random.

This finding has a broader implication for NN-augmented integrators in chaotic systems: single-trajectory global metrics (endpoint RMS or time-averaged RMS over $T \gg$ Lyapunov time) are unreliable measures of correction quality. The appropriate metric for evaluating NN corrections in chaotic systems is the local window metric restricted to the encounter window, ideally combined with ensemble testing across perturbed ICs.

9.3 Why Zone 2 Was Not Addressed by NN

Zone 2 ($r < 0.05$ AU) is a temporal resolution problem: the gravitational encounter evolves faster than the macro-timestep can resolve. Adaptive sub-stepping is the mathematically correct, exact solution — no learning is needed. Zone 3 is a force integration accuracy problem where the softened Newtonian force introduces residual error even at correct temporal resolution. The Phase 1 work established that Zone 2 sub-stepping alone reduced worst-case frontier error by 105x (from 161.7 AU to 1.54 AU). Applying an NN to Zone 2 would ask it to replace exact physics that is already being handled correctly by deterministic sub-stepping.

9.4 Scope and Generalisability

The encounter NN is validated for IC1, pair (1,2), at $dt=0.08$ yr with a single encounter at $t=82.56$ yr. Multi-IC generalisation is future work. The $dt=0.06$ and $dt=0.10$ ensemble results (10/10 seeds, iso_exit +61% and +74%) confirm the model correctly learns encounter dynamics across three timestep values; deployment robustness at those dt s requires the post-encounter Lyapunov constraint to be satisfied, which depends on the specific IC and simulation horizon.

10. Conclusions and Future Work

We have demonstrated two neural network approaches for close-encounter correction in three-body leapfrog integration:

(1) The **per-step scalar correction** (Phase 3 Set 2) makes locally accurate force corrections but cannot systematically improve $T=100\text{yr}$ global trajectories due to Lyapunov amplification. The local-global dissociation — demonstrated at three dt values with opposite signs — establishes that single-trajectory global metrics are insufficient for evaluating NN corrections in chaotic systems.

(2) The **encounter-level surrogate NN** achieves 93.7% exit-state position improvement and 94.0% velocity improvement at $dt=0.08\text{yr}$, persisting over a 4-year physical window (+48.7%). Global final position improves by +11.9% at 1.91x speedup. Ensemble testing across 10 IC perturbations confirms robustness after training-data augmentation, eliminating catastrophic failures from -305% to -13%.

(3) The **binding constraint on optimal dt** is the post-encounter Lyapunov horizon: $(T - t_{\text{encounter}})/\lambda_L < 3$. $dt=0.08$ satisfies this for IC1 (2.9 Lyapunov times), while $dt=0.06$ and $dt=0.10$ do not (10-11 Lyapunov times). This quantitative criterion provides a system-independent rule for optimal dt selection.

Future directions include: (a) extending the encounter NN to multiple ICs through more diverse rollout-local training data; (b) multi-encounter deployment logic to safely handle systems with more than one Zone 3 encounter per simulation; (c) tightened energy gate ($\text{relE}_{\text{corr}} < \text{noNN drift}$) as a physics-motivated safety threshold replacing the current 20% threshold; (d) testing on IC3 and IC4 configurations to validate the Lyapunov criterion's generalisability.

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